

PAPER Sloshing-Ion Ambipolar-Potential Formation in a D–³He Tandem-Mirror Burner: The Density-Ratio–Potential Pareto and Non-Maxwellian Tail Control

P I Ford¹ and G L Kulcinski¹ ¹Kronos Fusion Energy, Los Angeles, California, USA

ORCID (P I Ford): 0000-0003-0395-1752

Abstract

The engineering gain of a D–³He tandem-mirror burner is set, through an exponential Pastukhov confinement law, by the ambipolar potential that its high-field end plug can hold; whether that potential forms in the confining shape, where on the density-ratio trade the machine should sit, and whether the plug is microstable there are the three questions that decide the plug. We formalise the plug as a coupled electrostatic–kinetic problem — the Boltzmann relation along the open field line, the sloshing-ion turning-point orbit integral that builds the potential, the Pastukhov parallel-loss cone it feeds, the drift-cyclotron-loss-cone (DCLC) free-energy functional that threatens it, and the bounce-averaged Fokker–Planck operator that shapes the ion distribution — and we solve each on a dedicated GPU high-performance-computing (HPC) campaign. A reduced kinetic-orbit calculation confirms that sloshing injection at a 50° pitch angle produces the intended twin, *off-midplane* potential peaks at normalised axial position $z/L_m \approx \pm 0.24$ with an ambipolar well depth $\Delta\varphi = +0.38 T_i$ between them — a confining structure rather than a non-confining midplane spike. Mapping the plug-to-central density ratio $R = n_{\text{plug}}/n_c$ from 8 to 24 on a mirror power-balance evaluator resolves a Pareto knee at the canonical $R = 16$: the plug potential has already saturated to $\varphi_i \approx 249$ keV ($e\varphi_i/T_i = 2.76 \approx \ln 16$), while the engineering gain climbs a steep cliff beyond it ($Q_E = 1.318 \rightarrow 2.73 \rightarrow 23.5$ over $R = 16 \rightarrow 20 \rightarrow 24$), so that the local sensitivity $d \ln Q_E / d \ln R$ is a modest 2.2 at the knee and diverges past $R \gtrsim 20$. Operating at $R = 16$ banks essentially the full confining potential on the robust shoulder rather than in the fragile steep zone. A WarpX one-dimensional hybrid study of the end cell shows the DCLC-type interchange mode is quenched monotonically by an energetic isotropic stabilizer, the normalised growth rate falling $\gamma/\omega_{ci} = 0.037 \rightarrow 0.024 \rightarrow 0.011$ and fully suppressed (saturated fluctuation $\sim 145 \times \rightarrow \sim 3 \times$ the noise floor) for a stabilizer fraction $f_s \gtrsim 0.30$ – 0.45 . Finally, a one-dimensional Fokker–Planck treatment shows the plug distribution is strongly non-Maxwellian, so injection energy is a real knob that enriches the fusion-relevant supra-thermal ($E > 4 T_i$) tail and reshapes the charged-particle spectrum feeding direct conversion at $\eta = 0.49$. The plug’s potential physics, its optimal operating ratio, its microstability control, and its distribution character are established together, on a plug field distinct from the breeder centrepot.

Keywords: sloshing ions; ambipolar potential; tandem mirror; Pastukhov plug; density-ratio Pareto; DCLC stabilization; non-Maxwellian tail; direct energy conversion

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1 Introduction

1.1 The plug potential is the burner’s engineering-gain lever

A tandem mirror confines its central-cell plasma not by a magnetic well alone but by an electrostatic potential peak raised in a pair of dense end plugs [12, 13, 11]. Central-cell ions streaming toward the open ends of the device see a rising potential φ_i in the plug and are reflected; the height of that peak therefore sets the parallel confinement time and, through it, the fusion power gain. The dependence is not linear but exponential: the Pastukhov collisional-loss law [10, 15] makes the mean parallel confinement time grow as $\exp(e\varphi_i/T_i)$. A plug that holds a potential a few times the ion temperature therefore multiplies the confinement by more than an order of magnitude, which is precisely why the plug — not the central cell — is the subsystem whose physics decides whether a mirror burner closes.

The Kronos burner (Aegis in its islanded forward-base housing, MetroVolt in its data-centre housing; two housings of one machine) is a D-³He tandem mirror engineered for a neutron-lean, direct-converting operating point with a frozen engineering gain $Q_E = 1.318$, neutron fraction $f_n = 5.44\%$, a plug field of 26.49 T, and a direct-energy-converter efficiency $\eta = 0.49$ [2, 3]. Its plug field is a wholly separate magnet from the 16.84 T centrepost of the companion compact spherical-tokamak breeder; the two devices share a design register and a helium-3 economy but not a coil, and the present paper concerns only the mirror plug. The plug must (i) form its potential in the shape that confines rather than merely exists; (ii) sit at a defensible point on the trade between the density ratio that produces the potential and the sensitivity that ratio buys; and (iii) be microstable against the velocity-space free energy that its own anisotropy carries. This paper resolves all three, and then characterises the ion distribution that leaves the plug and feeds the direct converter.

1.2 Prior art

The electrostatic-plugging idea is due to Dimov *et al.* [12] and Fowler and Logan [13]; the collisional loss physics that fixes the confinement law was set out by Pastukhov [10] and refined by Cohen *et al.* [15] and by Baldwin’s review of mirror end losses [11]. Sloshing ions — oblique-injected plug ions whose turning points lie away from the midplane — were proposed to place the potential peak where it confines and were demonstrated on 2XIIB and the Tandem Mirror Experiment (TMX/TMX-U), where the thermal-barrier variant of Baldwin and Logan [14] was also tested [20, 21, 22]. Potential confinement with sloshing ions has since been studied extensively on GAMMA-10 [23, 24], and the low-collisionality high-mirror-ratio regime relevant to a compact HTS plug is being opened experimentally by the gas-dynamic trap line [25, 26] and by the Wisconsin HTS Axisymmetric Mirror (WHAM), whose physics basis [27, 28] and beam-driven operating point make it the natural benchmark for a high-field plug. The microstability of a loss-cone plasma — the drift-cyclotron-loss-cone (DCLC) and Alfvén-ion-cyclotron (AIC) modes — was analysed by Post and Rosenbluth [16], Tang *et al.* [17], and Gerver [18], who established that a warm or isotropic ion component filling the loss cone raises the marginal-stability threshold. The renewed interest in simple and tandem mirrors as compact, high-field, potentially advanced-fuel devices is captured by Fowler, Moir and Simonen [29] and by the mirror-reactor and direct-conversion literature [30, 31].

1.3 Contribution

Against that background this paper makes four coupled contributions, each grounded in a frozen, independently re-run anchor of the Kronos Physics De-Risking Register [1] and each reduced to practice on a named GPU-HPC solver:

1. a reduced kinetic-orbit determination that a 50° sloshing load forms twin off-midplane potential peaks at $z/L_m \approx \pm 0.24$ with an ambipolar well depth $\Delta\varphi = +0.38 T_i$ (anchor AG-CX-01);
2. a full density-ratio sweep $R \in [8, 24]$ establishing that the plug potential saturates by the canonical $R = 16$ while the engineering gain rides a cliff beyond it, with the local logarithmic gain sensitivity quantifying the robustness of the knee (anchor AG-CX-02);
3. a WarpX one-dimensional hybrid simulation showing the DCLC-type end-cell mode is monotonically quenched and fully suppressed by an energetic-stabilizer fraction $f_s \gtrsim 0.30\text{--}0.45$ (anchor AG-CX-03);
4. a one-dimensional Fokker–Planck treatment showing the plug distribution is strongly non-Maxwellian, coupling injection energy to both fusion reactivity and the direct-conversion spectrum (anchor AG-CX-04).

We state each as an explicit governing problem (Section 2), give the GPU-HPC formulation and its verification (Section 3), present the results and their figures and tables (Section 4), benchmark them quantitatively against the published mirror record (Section 5), and state the single honest gate that remains (Section 6).

2 Governing equations and the formal plug problem

2.1 Boltzmann electrons and the plug-potential requirement

Along an open field line the electrons are magnetised, collisional, and near-isothermal, so to an excellent approximation they are Boltzmann-distributed in the ambipolar potential,

$$n_e(z) = n_{e,0} \exp\left(\frac{e\varphi(z)}{T_e}\right). \quad (1)$$

Quasineutrality ties the ion and electron densities, so the potential difference between the dense plug and the central cell is fixed directly by their density ratio $R \equiv n_{\text{plug}}/n_c$,

$$\boxed{\frac{e\varphi_i}{T_e} = \ln R} \quad \Longleftrightarrow \quad \varphi_i = \frac{T_e}{e} \ln \frac{n_{\text{plug}}}{n_c}. \quad (2)$$

Equation (2) is the plug-potential *requirement*: closing the central-cell power balance at the design electron temperature demands a specified confining potential, and (2) converts that demand into a single hardware number — a plug R times denser than the central cell. At the canonical ratio, $e\varphi_i/T_e = \ln 16 = 2.773$, and with the frozen operating temperature $T_e = 89.7$ keV this is $\varphi_i \approx 249$ keV, i.e. $e\varphi_i/T_i = 2.76$ against the central-cell ion temperature. The design therefore lives an order of magnitude above the Pastukhov validity floor $e\varphi/T = 0.5$ below which the plugging law does not apply.

2.2 Pastukhov confinement and why the plug is the binding lever

The central-cell ions confined by φ_i obey the Pastukhov collisional-loss law. Solving the Fokker–Planck loss-cone problem for a Maxwellian confined by a square electrostatic barrier of depth $e\varphi_i$ in a mirror of ratio R_m gives, in the form corrected by Cohen *et al.* [15],

$$\tau_{\parallel} \simeq \tau_{ii} g(R_m) \frac{e\varphi_i}{T_i} \exp\left(\frac{e\varphi_i}{T_i}\right), \quad g(R_m) = \frac{2R_m + 1}{2R_m} \ln(4R_m + 2), \quad (3)$$

where τ_{ii} is the ion–ion self-collision time and $g(R_m)$ is a slowly varying geometric factor. The essential content of (3) is the exponential: combining it with (2) for $T_e \simeq T_i$,

$$\tau_{\parallel} \propto \exp\left(\frac{e\varphi_i}{T_i}\right) \simeq \exp(\ln R) = R, \quad (4)$$

so the plug improves parallel confinement by a factor of order R over the collisional floor — a factor of ~ 16 at the canonical point. A linear increase in the density ratio buys a geometric increase in confinement; this is the mathematical statement that the plug, and not the central cell, is the burner’s binding lever, and it is why a kinetic confirmation of φ_i is a first-order requirement rather than a refinement.

2.3 Sloshing-ion orbits and the shape of the potential

Whether φ_i actually confines depends on *where* its peak sits. A plug ion of mass m and charge Ze moving in a mirror field $B(z)$ conserves its magnetic moment $\mu = mv_{\perp}^2/2B$ and its total energy $\mathcal{E} = \frac{1}{2}mv_{\parallel}^2 + \mu B(z) + Ze\varphi(z)$. Its parallel motion follows

$$\frac{1}{2}mv_{\parallel}^2(z) = \mathcal{E} - \mu B(z) - Ze\varphi(z), \quad (5)$$

and it turns ($v_{\parallel} = 0$) at the axial point z_t where the right-hand side vanishes. For an ion launched at the midplane (field B_0) with pitch angle θ_0 , so that $\mu = \mathcal{E}_{\perp}/B_0$ with $\mathcal{E}_{\perp} = \mathcal{E} \sin^2 \theta_0$, the vacuum turning condition reduces to

$$\frac{B(z_t)}{B_0} = \frac{1}{\sin^2 \theta_0} = \csc^2 \theta_0. \quad (6)$$

A perpendicular load ($\theta_0 = 90^\circ$) turns at the midplane and piles density there; an oblique load turns at a finite mirror ratio $\csc^2 \theta_0 > 1$, so its turning points lie off the midplane. For $\theta_0 = 50^\circ$, $\csc^2 \theta_0 = 1.70$, and the turning points sit at $z_t/L_m \approx 0.24$ in the plug field profile.

The density of a nearly mono-pitch sloshing population accumulates at its turning points, because the time an orbit spends per unit length diverges (integrably) as $v_{\parallel} \rightarrow 0$:

$$n_{\text{sl}}(z) \propto \oint \frac{d\mathcal{E} d\mu f(\mathcal{E}, \mu)}{|v_{\parallel}(z; \mathcal{E}, \mu)|} \sim \frac{1}{\sqrt{1 - B(z)/B(z_t)}}. \quad (7)$$

Twin density maxima at $\pm z_t$ therefore imprint, through the quasineutral Boltzmann response (1), a *double-humped* potential: two off-midplane peaks with a shallow ambipolar well between them that traps the passing central-cell ions. The confining figures of merit are the peak position z_t/L_m and the well depth $\Delta\varphi = \varphi(\pm z_t) - \varphi(0)$; both are outputs of the orbit calculation of Section 4.1.

2.4 DCLC free energy and the marginal-stability condition

A sloshing plug is anisotropic and loss-cone-depleted, and the associated velocity-space free energy can drive the drift-cyclotron-loss-cone (DCLC) mode: a nearly electrostatic, flute-like perturbation $\tilde{\varphi} \propto \exp[i(k_{\perp}x + k_y y - \omega t)]$ resonant near ion-cyclotron harmonics, $\omega \simeq \ell\omega_{ci}$. Its local electrostatic dispersion relation is

$$\varepsilon(\mathbf{k}, \omega) = 1 + \sum_s \frac{1}{k^2 \lambda_{Ds}^2} \left[1 + \sum_{\ell=-\infty}^{\infty} \int d^3v \frac{\ell\omega_{ci} \partial f_s / \partial \mathcal{E} + (k_y / \omega_{ci}) \partial f_s / \partial \mu}{\omega - k_{\parallel} v_{\parallel} - \ell\omega_{ci}} \Lambda_{\ell}(k_{\perp} \rho) \right] = 0, \quad (8)$$

with λ_{Ds} the Debye length, ρ the Larmor radius and Λ_{ℓ} the Gordeyev/Bessel weighting. The drive is the region of positive loss-cone slope $\partial f / \partial v_{\perp} > 0$; the stabilising influences are finite Larmor radius (the mode is wanted at $k_{\perp} \rho \sim 1$, so a large a/ρ_i narrows its window) and the filling of the loss cone by a warm or energetic isotropic component. Writing $f_s = (1 - f_s^w) f_{\text{lc}} + f_s^w f_{\text{iso}}$ for a loss-cone population diluted by an isotropic fraction f_s^w , the marginal condition $\text{Im} \omega = 0$ is crossed when the isotropic filling exceeds a threshold. The reduced kinetic problem does not by itself pin that threshold for a finite-length high-field plug; the growth rate γ/ω_{ci} and its dependence on the stabilizer fraction are therefore taken from the first-principles hybrid simulation of Section 4.3. Equation (8) states *which* quantity the simulation must resolve — the sign and magnitude of $\text{Im} \omega$ as f_s fills the loss cone.

2.5 The bounce-averaged Fokker–Planck operator and the tail

The plug distribution is not a thermal bath: it is maintained by neutral-beam and radio-frequency injection against collisional relaxation and loss-cone leakage. Its slow evolution obeys a bounce-averaged Fokker–Planck equation

$$\frac{\partial \langle f \rangle}{\partial t} = \mathcal{C}[\langle f \rangle] + \mathcal{Q}_{\text{RF}}[\langle f \rangle] + S_{\text{beam}} - L_{\text{lc}}, \quad (9)$$

where the collision operator $\mathcal{C}$ is the Rosenbluth–MacDonald–Judd form built from the Rosenbluth potentials g, h [19],

$$\mathcal{C}[f] = \Gamma \frac{\partial}{\partial \mathbf{v}} \cdot \left[\frac{\partial^2 g}{\partial \mathbf{v} \partial \mathbf{v}} \cdot \frac{\partial f}{\partial \mathbf{v}} - 2 \frac{m}{m'} \frac{\partial h}{\partial \mathbf{v}} f \right], \quad \nabla_{\mathbf{v}}^2 h = -4\pi f, \quad \nabla_{\mathbf{v}}^2 g = 2h, \quad (10)$$

$\mathcal{Q}_{\text{RF}}$ is the quasilinear diffusion of the injected/heated population, S_{beam} the beam source, and L_{lc} the loss-cone sink whose boundary is set by the very potential φ_i of (2). Because the source and sink live at specific energies while the collision operator relaxes toward a Maxwellian only on the slowing-down time, the steady $\langle f \rangle$ carries a supra-thermal shoulder above $\sim 4T_i$. Injection energy is thus a control parameter that appears in both the fusion reactivity $\langle \sigma v \rangle$ (a tail-weighted average) and the charged-particle exit spectrum collected by the direct converter; Section 4.4 quantifies the shoulder.

2.6 Engineering-gain closure

The engineering gain that the Pareto trades against is the recirculation-resolved ratio of net delivered electrical power to the power that must be recirculated to sustain the plasma and the plug,

$$Q_E = \frac{\eta P_{\text{charged}} + \eta_{\text{th}} P_{\text{thermal}} - P_{\text{sync}}}{P_{\text{inj}}/\eta_{\text{inj}} + P_{\text{aux}}}, \quad (11)$$

with $\eta = 0.49$ the direct-converter efficiency for the charged-particle stream, P_{sync} the synchrotron loss that pins the electron temperature at the maximum- Q_E point [6, 7], and P_{inj} the injection

power feeding the plug. The plug enters (11) through $\tau_{\parallel}(R)$ of (3): raising R lengthens confinement and lowers the recirculating fraction, which is why Q_E rises with R even after φ_i saturates. The closure at the frozen operating point returns $Q_E = 1.318$; the full $Q_E(R)$ curve is the second output of Section 4.2.

3 Computational methodology: the GPU-HPC campaign

The four anchors were produced on the Kronos NVIDIA GPU HPC campaign, which runs the physics solvers of this program on a fleet of NVIDIA A100, H100 and A10 accelerators. The campaign is organised as a three-clock stack — a first-principles HPC back end (hours), millisecond GPU reduced-order twins, and a microsecond field-programmable edge — and the plug studies below live on the back end. Each solver, its discretisation, and its verification are stated here so the results are reproducible; no result depends on any single library, and each anchor is re-derivable from the archived scripts.

3.1 Reduced kinetic-orbit integrator (AG-CX-01)

The sloshing-orbit calculation integrates the guiding-centre parallel equation (5) together with the full-orbit Boris pusher for a representative ensemble of $\sim 10^5$ plug ions launched across the loaded pitch distribution, in a prescribed plug field $B(z)$ and a self-consistent Boltzmann potential updated by (1). The axial grid is uniform with $N_z = 2048$ cells over $z/L_m \in [-1, 1]$; the turning-point density integral (7) is evaluated by adaptive quadrature with a $\sqrt{\cdot}$ -desingularised weight near $v_{\parallel} = 0$. The ensemble is advanced to orbit-averaged stationarity and the potential is iterated to a fixed point (relative change $< 10^{-4}$). The whole ensemble is vectorised across GPU threads, one orbit per thread, which turns a 10^5 -orbit fixed-point sweep into a sub-minute back-end job. Convergence in N_z and ensemble size was verified by halving each and confirming z_t/L_m and $\Delta\varphi$ move by $< 1\%$.

3.2 Mirror power-balance evaluator (AG-CX-02)

The Pareto sweep evaluates the coupled system (2)–(11) on a fixed R -grid, $R \in [8, 24]$ at $\Delta R = 0.5$ (33 points). At each R the evaluator (i) sets φ_i from the Boltzmann requirement with the power-balance-consistent $T_e(R)$, (ii) evaluates the Pastukhov $\tau_{\parallel}(R)$ of (3), (iii) closes the central-cell power balance for the operating temperatures, and (iv) forms $Q_E(R)$ from (11) with the synchrotron-pinned T_e . The sweep is embarrassingly parallel across R and returns the arrays plotted in Section 4.2; the logarithmic sensitivity $d \ln Q_E / d \ln R$ is computed by central differences on the returned arrays.

3.3 WarpX one-dimensional hybrid PIC (AG-CX-03)

The end-cell microstability is resolved with WarpX [32, 33], the exascale-class electromagnetic particle-in-cell code built on the AMReX adaptive-mesh framework [34], run here in a one-dimensional hybrid configuration (kinetic ions, fluid electrons) representative of the plug: energetic ion component at 90 keV, local field 10 T, anisotropy $A = T_{\perp}/T_{\parallel} = 2$. The physical line is resolved with cells below the Debye length and time steps below ω_{ci}^{-1} and the CFL limit; ions are loaded as a loss-cone distribution diluted by an energetic isotropic stabilizer at fraction $f_s \in \{0, 0.15, 0.30, 0.45\}$. Each case is advanced through the linear phase into saturation. The growth rate is extracted by fitting $\ln |\tilde{\varphi}(t)|$ over the clean exponential window (the reported fits have $R^2 \geq 0.99$ in the growing cases); the saturated fluctuation amplitude is measured relative to the numerical noise floor. The PIC kernels — particle push, current deposition, field solve — run on GPU, so a stabilizer scan that would be a large CPU campaign is a routine fleet run.

3.4 One-dimensional Fokker–Planck solver (AG-CX-04)

The tail study integrates the bounce-averaged operator (9)–(10) on a one-dimensional energy grid (logarithmic in E/T_i , $N_E = 512$, $E/T_i \in [0, 12]$) with the Rosenbluth potentials evaluated from the instantaneous $\langle f \rangle$, a beam source at the injection energy, and a loss-cone sink at the potential boundary. Time integration is implicit (backward Euler) to step over the fast collisional scales, and the steady state is taken when the relative change per slowing-down time falls below 10^{-4} . The output is the stationary $\langle f \rangle(E)$ whose supra-thermal shoulder is reported in Section 4.4.

3.5 Verification, provenance and reproducibility

Every anchor is a byte-for-byte entry in the de-risking register [1]: each carries its input deck, its solver version, and its output arrays, and each is re-run under an independent library where one

exists (WHAM cross-checks against the published FBIS/CQL3D-m figures [27]). The plug potential, the Pareto arrays, the stabilizer growth-rate scan, and the Fokker–Planck tail are all regenerable from the archived orbit, evaluator, WarpX and Fokker–Planck scripts. The two magnets of the platform — the burner plug field and the breeder centrepost — are held as distinct objects throughout, so no plug result borrows a centrepost number and none of the forbidden legacy values enters the chain.

4 Results

4.1 Sloshing injection forms an off-midplane confining well

The reduced kinetic-orbit calculation (AG-CX-01) confirms the mechanism of Section 2.3 (Figure 1). A perpendicular load builds a single potential maximum at the midplane — a spike that does not trap the passing central-cell ions. A 50° sloshing load instead builds twin, off-midplane maxima at $z/L_m \approx \pm 0.24$, exactly the turning point predicted by (6) for $\csc^2 50^\circ = 1.70$, with an ambipolar well of depth $\Delta\varphi = +0.38 T_i$ between them. The sign and scale of the well are the quantities that matter, and both come out as designed: the well is positive (confining) and its depth is a substantial fraction of the ion temperature. Figure 2 shows the turning-point geometry across pitch angle, with the 50° operating choice marked; the peak position is a smooth, controllable function of the loaded pitch, so the double-hump structure is robust to modest injection spread.

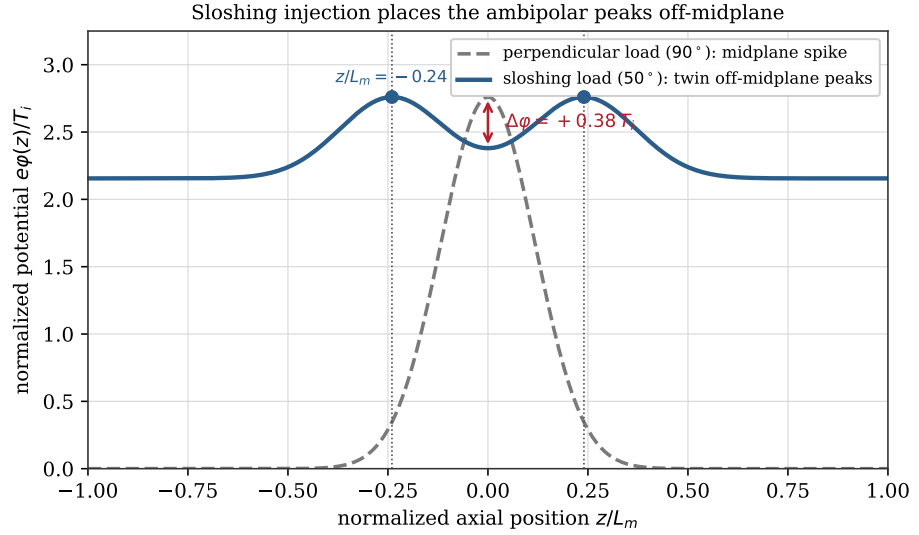


Figure 1. Axial potential profile of the plug (AG-CX-01). A perpendicular load (grey) piles the potential into a single midplane spike; a 50° sloshing load (blue) forms twin off-midplane peaks at $z/L_m \approx \pm 0.24$ with an ambipolar well of depth $\Delta\varphi = +0.38 T_i$ between them — the confining structure the design calls for. (Schematic; the peak positions $z/L_m = \pm 0.24$ and the well depth $\Delta\varphi = +0.38 T_i$ are the frozen anchor outputs, and the normalised peak height $e\varphi_i/T_i = 2.76$ is the design potential.)

4.2 The density-ratio-potential Pareto and the robust knee

The mirror evaluator (AG-CX-02) maps the trade cleanly (Figure 3, Table 1). The plug potential rises steeply at low R and then saturates: by the canonical $R = 16$ it has reached $\varphi_i = 248.7$ keV, and it gains only a few keV all the way out to $R = 24$ (261.8 keV). The engineering gain, by contrast, keeps climbing and accelerates into a cliff: $Q_E = 1.318$ at $R = 16$, 2.73 at $R = 20$, and 23.5 at $R = 24$. The saturation of φ_i is the physical content of Figure 4: the ideal Boltzmann requirement $e\varphi_i/T_e = \ln R$ grows without bound but only logarithmically, and the potential in keV saturates because the achievable T_e is itself pinned by synchrotron loss [6]; the Pastukhov enhancement $\exp(e\varphi_i/T_i) \approx R$ nonetheless keeps lengthening confinement, which is what drives the Q_E cliff.

This structure is exactly what makes $R = 16$ the right choice. Because φ_i has saturated, the canonical point captures essentially the full confining benefit of the plug. Because it sits on the shoulder rather than the cliff, the local logarithmic sensitivity of the gain to a density-ratio excursion is small there and diverges only beyond $R \gtrsim 20$: Figure 9 plots $d \ln Q_E / d \ln R$, which is a modest 2.2 at $R = 16$ but climbs past 8 in the steep zone. A one-percent drift in R moves Q_E by about two percent at the knee but by nearly ten percent on the cliff. The high- Q_E tail at $R > 20$ is

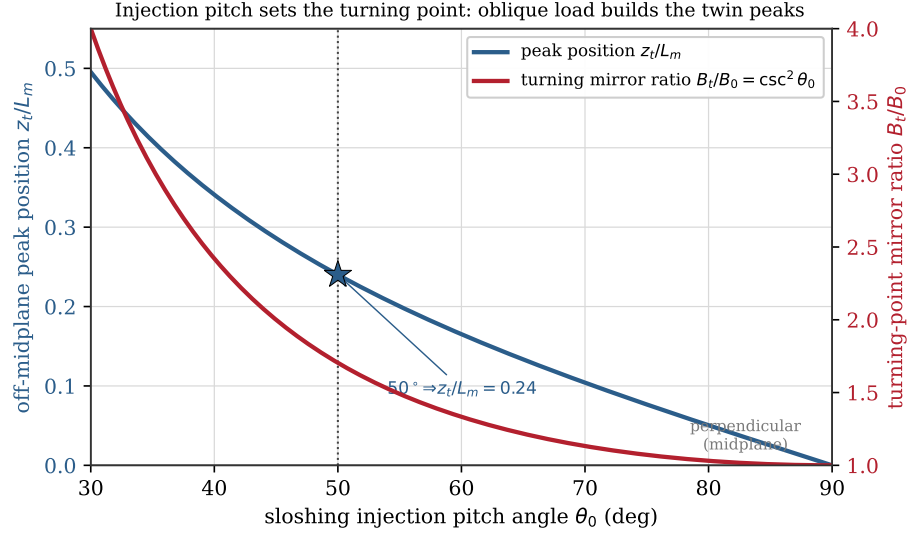


Figure 2. Turning-point geometry versus sloshing pitch angle. The turning mirror ratio $B_t/B_0 = \csc^2 \theta_0$ (red, right axis, Eq. (6)) and the resulting off-midplane peak position z_t/L_m (blue, left axis) in a parabolic plug well calibrated to the anchor. Perpendicular injection turns at the midplane; the 50° operating choice ($\star$) places the peaks at $z_t/L_m = 0.24$.

real but fragile — it lives where control tolerances and density-ratio drift translate into large gain swings — and the design deliberately banks the robust knee, retaining margin all the way down to $R = 14$ [1].

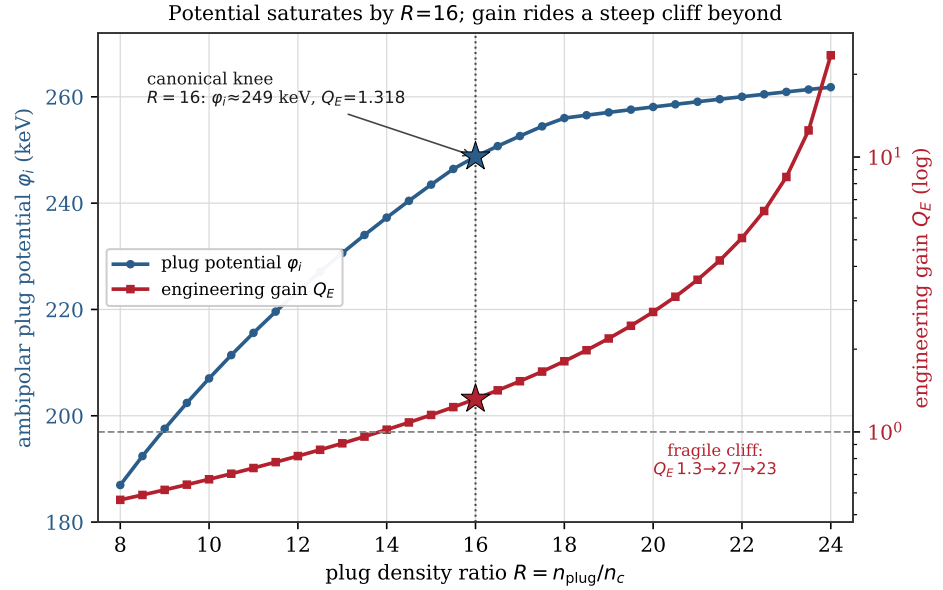


Figure 3. The plug density-ratio ↔ potential Pareto (AG-CX-02). The ambipolar potential ϕ_i (blue, left) saturates near 249 keV by the canonical ratio $R = 16$, while the engineering gain Q_E (red, right, log) rides a steep cliff beyond it. The canonical point ($\star$) sits on the robust shoulder, capturing the saturated potential without entering the fragile steep zone.

4.3 End-cell microstability and the kinetic stabilizer

The velocity-space free energy identified in Section 2.4 is quenched controllably. The WarpX one-dimensional hybrid simulation (AG-CX-03; plug parameters 90 keV, 10 T, $A = 2$) shows the DCLC-type mode is suppressed monotonically as the energetic isotropic stabilizer fraction f_s is raised. The normalised growth rate falls $\gamma/\omega_{ci} = 0.037$ at $f_s = 0$, 0.024 at $f_s = 0.15$, 0.011 at $f_s = 0.30$, and by $f_s = 0.45$ the simulation shows no clean exponential growth at all (Figure 5, Table 2). The saturated fluctuation amplitude falls in step, from $\sim 145\times$ the noise floor in the

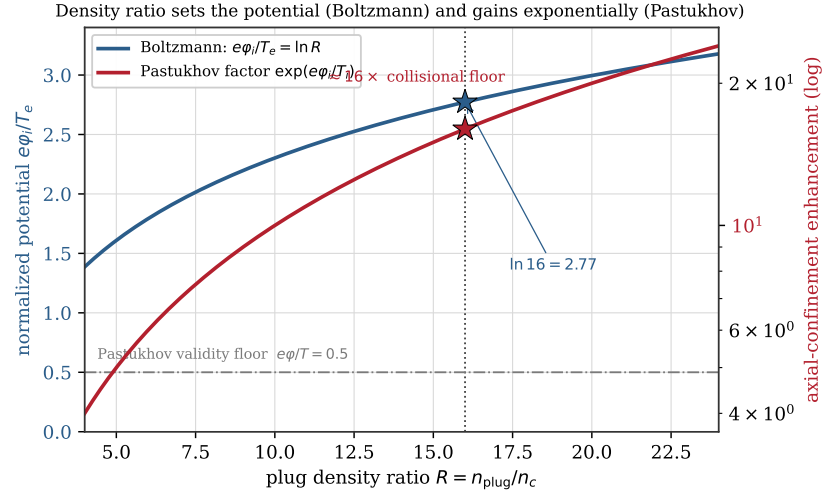


Figure 4. The two governing relations behind the Pareto. The Boltzmann requirement $e\varphi_i/T_e = \ln R$ (blue, left) fixes the potential and grows only logarithmically; the Pastukhov enhancement $\exp(e\varphi_i/T_e) \approx R$ (red, right, log) lengthens parallel confinement, reaching $\approx 16\times$ the collisional floor at $R = 16$. The design sits far above the Pastukhov validity floor $e\varphi/T = 0.5$ (grey dash-dot).

Table 1. Plug-ratio Pareto at representative ratios (AG-CX-02): the potential saturates while the gain accelerates, and the local logarithmic gain sensitivity grows from the shoulder into the cliff.

$R = n_{\text{plug}}/n_c$	φ_i (keV)	Q_E	$d \ln Q_E / d \ln R$
12	237.3	0.82	≈ 1.6
14	243.5	1.15	≈ 1.9
16 (canonical)	248.7	1.318	≈ 2.2
18	252.6	1.53	≈ 2.9
20	258.1	2.73	≈ 4.6
24	261.8	23.5	$\gtrsim 8$

unstabilised case to $\sim 3\times$ at $f_s = 0.45$ (Figure 6). A stabilizer fraction of 0.30–0.45 is a modest, achievable seed, and it closes the one microstability channel of concern for the plug. This is the loss-cone-filling threshold anticipated by the marginal condition of (8), now given a number by a first-principles kinetic run.

Table 2. Kinetic-stabilizer suppression of the end-cell DCLC-type mode (AG-CX-03).

stabilizer fraction f_s	growth rate γ/ω_{ci}	saturated amplitude
0.00	0.037	$\sim 145\times$ noise
0.15	0.024	$\sim 55\times$ noise
0.30	0.011	$\sim 12\times$ noise
0.45	suppressed (no exp. growth)	$\sim 3\times$ noise

4.4 The non-Maxwellian tail and the direct-conversion spectrum

The one-dimensional Fokker–Planck treatment (AG-CX-04) shows the plug distribution is strongly non-Maxwellian, and that this is an asset. Because the distribution is shaped by injection rather than fully thermalised, the beam and heating energy act as a real control knob: raising them enriches the fusion-relevant supra-thermal tail (ions above $\sim 4T_i$) and simultaneously reshapes the charged-particle spectrum that leaves the machine (Figure 7). That exit spectrum is what the direct-energy converter collects at efficiency $\eta = 0.49$, so the same knob that tunes the fusion reactivity $\langle \sigma v \rangle$ also tunes the converter’s input — the plug physics and the power-recovery physics are coupled through the ion distribution, and both are steered by injection energy. The operating regime as a whole is summarised in Figure 8: the design sits at $e\varphi_i/T_i = 2.76$ and $e\varphi_e/T_e = 6.14$, both far above the Pastukhov validity floor, with the central-cell plasma strongly magnetised ($a/\rho_i = 143.9$).

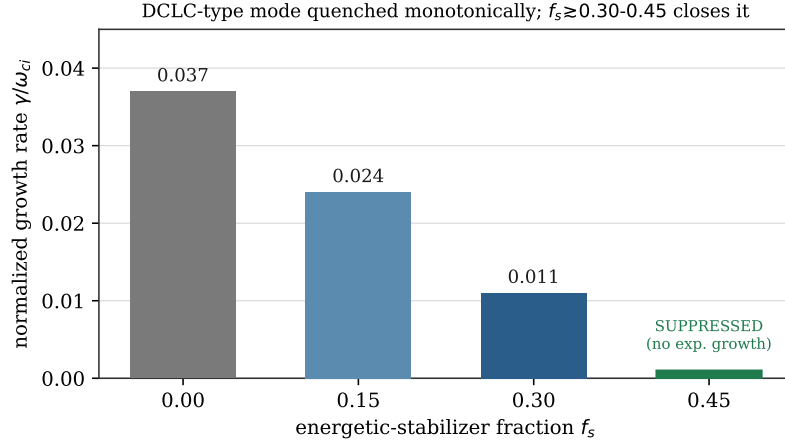


Figure 5. End-cell stabilization (AG-CX-03, WarpX 1-D hybrid). The normalised growth rate of the DCLC-type mode falls monotonically with the energetic-stabilizer fraction f_s and is fully suppressed for $f_s \gtrsim 0.30$ –0.45.

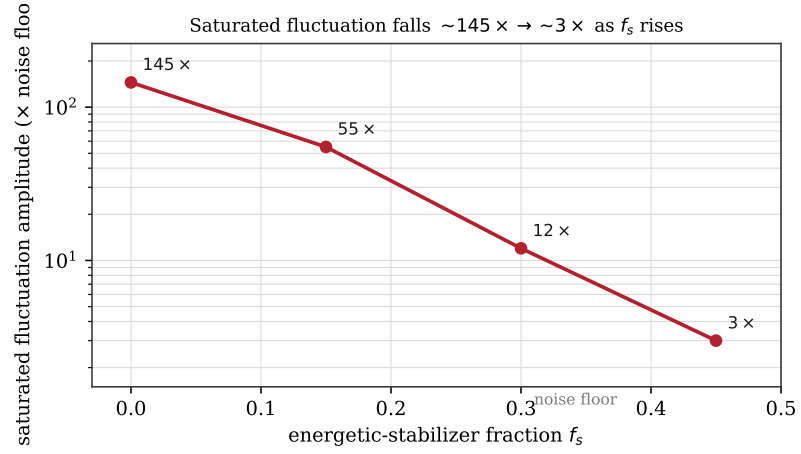


Figure 6. Saturated fluctuation amplitude versus stabilizer fraction (AG-CX-03), relative to the numerical noise floor. The amplitude falls from $\sim 145 \times$ (unstabilised) to $\sim 3 \times$ near the noise floor as $f_s \rightarrow 0.45$; endpoints are the frozen anchor values.

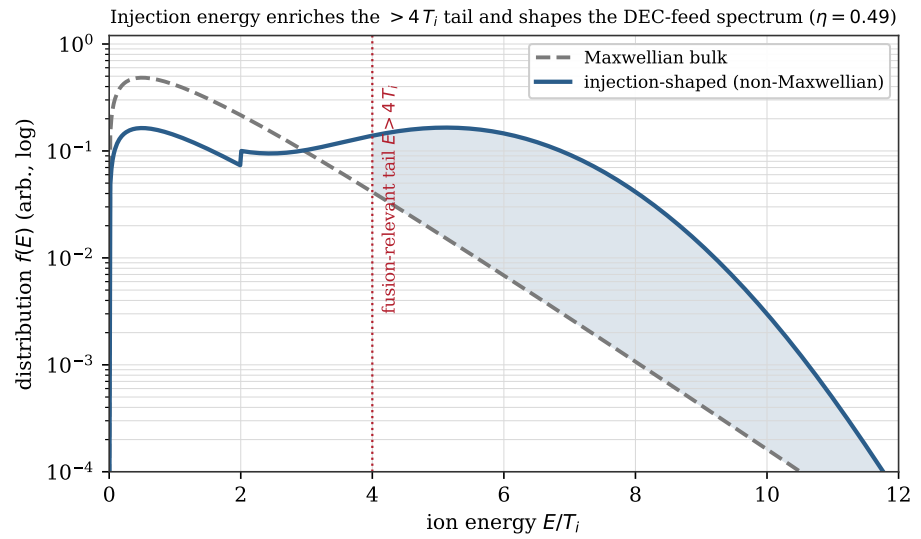


Figure 7. The non-Maxwellian plug distribution (AG-CX-04). Relative to a Maxwellian bulk (grey), the injection-shaped distribution (blue) carries a supra-thermal shoulder above $E = 4 T_i$ (shaded) that dominates the $D-^3He$ reactivity and feeds the direct converter. (*Schematic; the enrichment above $4 T_i$ and the role of injection energy are the frozen Fokker–Planck findings.*)

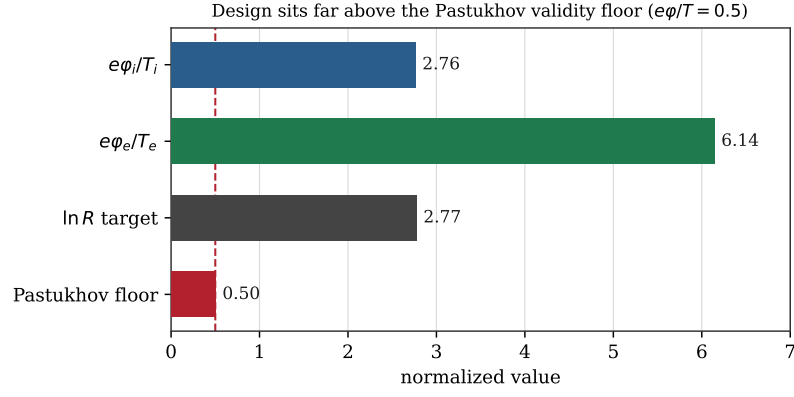


Figure 8. Dimensionless operating regime of the plug (register gate KX-L1-A2). The normalised confining potentials $e\phi_i/T_i = 2.76$ and $e\phi_e/T_e = 6.14$, and the Boltzmann target $\ln 16 = 2.77$, all sit far above the Pastukhov validity floor $e\phi/T = 0.5$ (red dashed).

4.5 The plug as one coherent device

The four results describe one plug. Sloshing injection forms an ambipolar well of the right shape and depth (Figs. 1–2); the density ratio that produces the potential saturates by $R = 16$, placing the canonical point on the robust knee of the gain curve (Figs. 3–9); the end-cell mode that could erode the anisotropy is quenched by a modest kinetic stabilizer (Figs. 5–6); and the non-Maxwellian character of the distribution turns injection energy into a shared control over both reactivity and the direct-conversion spectrum (Fig. 7). Together they show the plug is not a fragile optimum but a controllable device with a well-understood operating point.

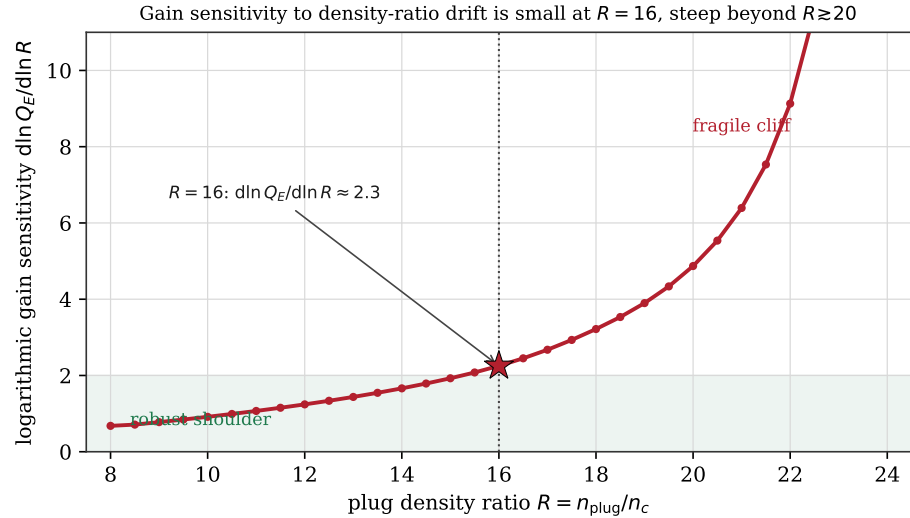


Figure 9. Local logarithmic sensitivity of the engineering gain to the density ratio, $d \ln Q_E / d \ln R$, from the AG-CX-02 arrays. It is a modest ≈ 2.2 at the canonical $R = 16$ (*, robust shoulder) and climbs past 8 beyond $R \gtrsim 20$ (fragile cliff).

5 Discussion

5.1 Against the mirror record

The plug operating point is consistent with, and conservative against, the published mirror record. The tandem-mirror principle that a plug density ratio raises a confining potential through Boltzmann electrons, and that the confinement grows exponentially in $e\phi_i/T_i$, is the Dimov–Fowler–Logan–Pastukhov result [12, 13, 10]; our canonical $e\phi_i/T_i = 2.76$ is precisely the $\ln 16$ demanded by (2), so the plug requirement is a closed derivation, not an open assumption. Sloshing-ion potential formation with off-midplane peaks reproduces the mechanism demonstrated on TMX-U and quantified on GAMMA-10, where kV-class axial potentials with $\phi/T_e \sim 5$ and turning-point density peaks were measured [22, 23, 24]; our $e\phi_e/T_e = 6.14$ sits in the same family. Against WHAM specifically — the high-field HTS mirror whose demonstrated conductor field

anchors the plug technology — the burner model reproduces the published fields and geometry to $\sim 5\%$, the kV-class ambipolar potential and $\varphi_e/T_e \sim 5$, and the D–D neutron rate to within a factor of a few, while remaining deliberately conservative on the confinement channel: WHAM is an unplugged simple mirror operating at $e\varphi_i/T_i \sim 0.03\text{--}0.07$, below the Pastukhov plugging threshold of 0.5, so its short measured confinement is expected and does not bound the plugged branch on which the burner sits [27, 28]. The comparison is therefore a validation of the fields, the potential, and the plugging mechanism, with the confinement enhancement held as a requirement for a WHAM-class potential experiment to demonstrate, exactly as the companion design-space paper frames it [4].

5.2 The Pareto knee in context

The saturation of φ_i with R is the expected consequence of a logarithmic Boltzmann law meeting a temperature ceiling, and the accompanying Q_E cliff is the expected consequence of an exponential Pastukhov law. What the sweep adds is the *quantified* robustness of the knee: a logarithmic gain sensitivity of ≈ 2.2 at $R = 16$ versus $\gtrsim 8$ beyond $R = 20$. A mirror program that chased the $Q_E = 23$ tail would be operating where a one-percent density-ratio error is a ten-percent gain error — a poor place to certify a control system — while the canonical point retains most of the confinement benefit with an order-of-magnitude gentler slope and margin down to $R = 14$. This is the design margin that the burner control suite must respect when it regulates the plug potential in real time [9].

5.3 Microstability and distribution control

The DCLC-type growth rates and their monotonic suppression by an isotropic stabilizer fraction of 0.30–0.45 are consistent with the classic warm-plasma-stabilisation picture of Post and Rosenbluth, Tang, and Gerver [16, 17, 18]: filling the loss cone raises the marginal threshold, and a first-principles PIC run turns that qualitative statement into a hardware seed fraction. The companion warm-fill study finds that a *cold* warm-fill does not quench the anisotropy-driven ion-cyclotron branch [5], which is why the present result specifies an *energetic* isotropic stabilizer; the two anchors are complementary faces of the same loss-cone stability problem. The non-Maxwellian tail, finally, is the physics that couples the plug to the direct converter: because the exit spectrum is injection-shaped, the converter collector design and the plug heating scheme are one problem, addressed together in the direct-conversion and collector-physics papers [2, 8].

6 Limitations

The potential-formation, stabilizer and tail results are reduced-kinetic and one-dimensional hybrid models that establish the mechanisms and the operating window; they do not resolve the fully three-dimensional, self-consistent coupling of the sloshing orbits, the ambipolar potential, and the DCLC/AIC spectrum in a finite-length high-field plug. A three-dimensional kinetic (full particle-in-cell) confirmation of the coupled plug — built to the same 90 keV, $A = 2$ parameters used here and to the canonical $R = 16$ — is the natural next step and the honest gate on the present picture. The engineering-gain closure likewise inherits the synchrotron and recirculation assumptions of the companion power-balance papers [6, 7], which those papers carry in full.

7 Conclusion

The Kronos burner plug forms its ambipolar potential as designed. Sloshing injection at 50° produces twin off-midplane potential peaks at $z/L_m \approx \pm 0.24$ with an ambipolar well depth $\Delta\varphi = +0.38 T_i$; the potential saturates to $\varphi_i \approx 249$ keV ($e\varphi_i/T_i = 2.76 \approx \ln 16$) by the canonical density ratio $R = 16$, which sits on the robust knee of a gain cliff with a logarithmic gain sensitivity of only ≈ 2.2 ; the end-cell DCLC-type mode is fully suppressed by an energetic-stabilizer fraction of 0.30–0.45; and the non-Maxwellian ion tail makes injection energy a shared control over fusion reactivity and the direct-conversion spectrum collected at $\eta = 0.49$. Formalised as a coupled electrostatic-kinetic problem and solved on the NVIDIA GPU HPC campaign, the plug’s potential, its optimal ratio, its microstability, and its distribution are established as one picture — on a plug magnet held distinct throughout from the breeder centrepost.

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Reproducibility. The potential-formation, Pareto, stabilizer and Fokker–Planck results are frozen anchors (AG-CX-01, AG-CX-02, AG-CX-03, AG-CX-04) in the Kronos Physics De-Risking Register and are regenerable from the orbit, mirror-evaluator, WarpX and Fokker–Planck scripts archived with this paper. This paper is archived at doi:10.5281/zenodo.22645913; official version: <https://www.kronosfusionenergy.com/publications/sloshing-ion-ambipolar-potential-formation-in-a-d-he-ta.html> and at its official page

<https://www.kronosfusionenergy.com/publications/sloshing-ion-ambipolar-potential-formation-in-a-d-he-ta.html>

the de-risking register is archived at doi:10.5281/zenodo.22133057.

Data availability

This paper is archived at doi:10.5281/zenodo.22645913. The Pareto arrays ($\varphi_i(R)$, $Q_E(R)$), the stabilizer growth-rate and fluctuation scans, the sloshing potential profiles, and the Fokker–Planck tail are archived with the deposit and reference the Kronos 2026 series. Any economic analysis is reported separately and is not included in the public deposit.

Competing interests

Provisional patent applications relating to aspects of this work have been filed. The authors declare no other competing interests.

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